

ISC Class XII Physics

Chapter 7 Alternating Current

57 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. Mean, Peak and rms Values of AC
- II. AC Circuit with either of R, L or C
- III. AC Circuit containing R and L
- IV. AC Circuit containing R and C
- V. L-C-R Circuit
- VI. L-C Circuits and Oscillations
- VII. Power Consumed in AC Circuits
- VIII. Transformer

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Formula Sheet

I. Mean, Peak and rms Values of AC

rms and mean value (sinusoidal AC)

$$I_{rms} = \frac{I_0}{\sqrt{2}}, \quad V_{rms} = \frac{V_0}{\sqrt{2}}, \quad I_{mean} = \frac{2I_0}{\pi}$$

I_0, V_0 = peak current/voltage; mean value is defined only over a half-cycle (it is zero over a full cycle).

rms value of a sine + cosine combination

$$I = A \sin \omega t + B \cos \omega t \Rightarrow I_0 = \sqrt{A^2 + B^2}, \quad I_{rms} = \frac{I_0}{\sqrt{2}}$$

A, B = amplitudes of the sine and cosine components (same frequency).

rms value with a DC offset

$$I = I_{dc} + I_{ac} \sin \omega t \Rightarrow I_{rms} = \sqrt{I_{dc}^2 + \frac{I_{ac}^2}{2}}$$

I_{dc} = steady (DC) part, I_{ac} = peak value of the varying part.

II. AC Circuit with either of R, L or C

Inductive and capacitive reactance

$$X_L = \omega L = 2\pi f L, \quad X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

X_L = inductive reactance, X_C = capacitive reactance, f = frequency, L = inductance, C = capacitance.

Phase relations for pure elements

Pure R : V, I in phase. Pure L :

V leads I by 90° . Pure C : I leads V by 90°

Governs the phase relation between voltage and current for a single circuit element.

III. AC Circuit containing R and L

Impedance and phase angle (R-L series)

$$Z = \sqrt{R^2 + X_L^2}, \quad \tan \phi = \frac{X_L}{R}, \quad I = \frac{V}{Z}$$

Z = impedance, ϕ = phase angle between V and I (current lags voltage).

IV. AC Circuit containing R and C**Impedance and phase angle (R-C series)**

$$Z = \sqrt{R^2 + X_C^2}, \quad \tan \phi = \frac{X_C}{R}$$

Same impedance triangle as R - L , but with I leading V instead of lagging.

V. L-C-R Circuit**Impedance and phase angle (series L-C-R)**

$$Z = \sqrt{R^2 + (X_L - X_C)^2}, \quad \tan \phi = \frac{X_L - X_C}{R}$$

If $X_L > X_C$ the circuit is inductive (current lags); if $X_C > X_L$ it is capacitive (current leads).

Resonance condition

$$X_L = X_C \Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}, \quad Z_{min} = R$$

f_0 = resonant frequency; at resonance the circuit is purely resistive and current is maximum.

Quality factor

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Q = quality factor, a measure of the sharpness of resonance.

VI. L-C Circuits and Oscillations**Series L-C circuit**

$$V = |V_L - V_C|, \quad Z = |X_L - X_C|$$

In a pure L - C series circuit, V_L and V_C are 180° out of phase, so they subtract, not add.

Minimum-impedance / maximum-current condition

$$\omega L = \frac{1}{\omega C}$$

Used to find the L or C needed to minimise impedance (maximum brightness/current).

VII. Power Consumed in AC Circuits

Average power and power factor

$$P = V_{rms} I_{rms} \cos \phi, \quad \cos \phi = \frac{R}{Z}$$

For a pure resistor $\phi = 0$, so $P = I_{rms}^2 R = V_{rms} I_{rms}$.

VIII. Transformer

Ideal transformer relation

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}, \quad \eta = \frac{P_{out}}{P_{in}} \times 100\%$$

N_p, N_s = primary/secondary turns; V_p, V_s, I_p, I_s = primary/secondary voltage and current;
 η = efficiency.

I. Mean, Peak and rms Values of AC

Question 1

An alternating current at any instant is given by $I = (3 + \sqrt{32} \sin 100\pi t)$ A. What is the rms value of this current?

Given: $I = 3 + \sqrt{32} \sin 100\pi t$ (A), i.e. a DC part $I_{dc} = 3$ A and an AC part of peak value $\sqrt{32}$ A

Formula: $I_{rms} = \sqrt{I_{dc}^2 + \frac{I_{ac}^2}{2}}$

Solution:

Here $I_{dc} = 3$ A and peak of AC term $I_{ac} = \sqrt{32}$ A, so $I_{ac}^2 = 32$.

$$I_{rms} = \sqrt{3^2 + \frac{32}{2}} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ A}$$

Final Answer

$$I_{rms} = 5 \text{ A}$$

Common Mistake: Do not simply add $3 + \sqrt{32}/\sqrt{2}$; the DC part and AC part must be combined as $\sqrt{I_{dc}^2 + I_{ac}^2/2}$, not added directly.

Question 2

An AC current is given by $I = (3 \sin \omega t + 4 \cos \omega t)$ A. Find its rms value.

Given: $I = 3 \sin \omega t + 4 \cos \omega t$ (A)

Formula: $I_0 = \sqrt{A^2 + B^2}, \quad I_{rms} = \frac{I_0}{\sqrt{2}}$

Solution:

Since $\sin \omega t$ and $\cos \omega t$ have the same frequency, they combine into a single sinusoid of peak value:

$$I_0 = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ A}$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} = \frac{5}{\sqrt{2}} = \frac{5\sqrt{2}}{2} \text{ A}$$

Final Answer

$$I_{rms} = \frac{5\sqrt{2}}{2} \text{ A} \approx 3.54 \text{ A}$$

Common Mistake: Do not add the individual rms values of $3 \sin \omega t$ and $4 \cos \omega t$ separately; combine amplitudes first using $\sqrt{A^2 + B^2}$, then take rms.

Question 3

An alternating current is given by $I = 50 \sin 100\pi t$. Find the frequency of current, mean current during a positive half-cycle, virtual (rms) current, and the current at $t = (1/300) \text{ s}$.

★ Likely Exam Question

Given: $I = 50 \sin 100\pi t$ (A); compare with $I = I_0 \sin \omega t$, so $I_0 = 50 \text{ A}$, $\omega = 100\pi \text{ rad/s}$

Formula: $f = \frac{\omega}{2\pi}$, $I_{mean} = \frac{2I_0}{\pi}$, $I_{rms} = \frac{I_0}{\sqrt{2}}$, $I(t) = I_0 \sin \omega t$

Solution:

Frequency:

$$f = \frac{\omega}{2\pi} = \frac{100\pi}{2\pi} = 50 \text{ Hz}$$

Mean current (positive half cycle):

$$I_{mean} = \frac{2I_0}{\pi} = \frac{2 \times 50}{\pi} = \frac{100}{\pi} = 31.8 \text{ A}$$

Virtual (rms) current:

$$I_{rms} = \frac{I_0}{\sqrt{2}} = \frac{50}{\sqrt{2}} = 35.4 \text{ A}$$

Current at $t = 1/300 \text{ s}$:

$$I = 50 \sin \left(100\pi \times \frac{1}{300} \right) = 50 \sin \left(\frac{\pi}{3} \right) = 50 \times 0.866 = 43.3 \text{ A}$$

Final Answer

$$f = 50 \text{ Hz}, I_{mean} = 31.8 \text{ A}, I_{rms} = 35.4 \text{ A}, I(t=1/300 \text{ s}) = 43.3 \text{ A}$$

Common Mistake: The mean-value formula $2I_0/\pi$ applies only to a half cycle; the mean current over a full cycle is always zero.

Question 4

A hot-wire ammeter in an AC circuit reads 10 A (rms value). What is the peak current?

Given: $I_{rms} = 10 \text{ A}$

Formula: $I_0 = I_{rms} \times \sqrt{2}$

Solution:

$$I_0 = 10 \times \sqrt{2} = 14.14 \text{ A}$$

Final Answer

$$I_0 \approx 14.1 \text{ A}$$

Common Mistake: A hot-wire (thermal) ammeter always reads the rms value, never the peak value directly.

Question 5

A 10Ω resistance is connected across 220 V – 50 Hz AC supply. Find the time taken by the current to change from its maximum value to the rms value.

Given: $R = 10 \Omega$, $f = 50 \text{ Hz}$

Formula: $I = I_0 \cos \omega t$, where $I = \frac{I_0}{\sqrt{2}}$

Solution:

Taking $t = 0$ at the instant of maximum current, $I = I_0 \cos \omega t$. Setting $I = I_0/\sqrt{2}$:

$$\cos \omega t = \frac{1}{\sqrt{2}} \Rightarrow \omega t = \frac{\pi}{4}$$

$$t = \frac{\pi}{4\omega} = \frac{\pi}{4(2\pi f)} = \frac{1}{8f} = \frac{1}{8 \times 50} = \frac{1}{400} \text{ s} = 2.5 \times 10^{-3} \text{ s}$$

Final Answer

$$t = 2.5 \text{ ms}$$

Common Mistake: Value of resistance R does not affect this time; students often try to bring R into the calculation unnecessarily.

Question 6

If the peak alternating current in a circuit is 21.2 A, find the reading of an ammeter connected in the circuit.

Given: $I_0 = 21.2$ A

Formula: $I_{rms} = \frac{I_0}{\sqrt{2}}$

Solution:

$$I_{rms} = \frac{21.2}{\sqrt{2}} = 15.0 \text{ A}$$

Final Answer

Ammeter reading = 15 A

Common Mistake: An AC ammeter always reads rms value, not the peak value 21.2 A directly.

Question 7

The equation of an AC voltage is $V = 100\sqrt{2} \sin(100\pi t)$. Find the rms value and the frequency of the voltage.

Given: $V = 100\sqrt{2} \sin(100\pi t)$; $V_0 = 100\sqrt{2}$ V, $\omega = 100\pi$ rad/s

Formula: $V_{rms} = \frac{V_0}{\sqrt{2}}$, $f = \frac{\omega}{2\pi}$

Solution:

$$V_{rms} = \frac{100\sqrt{2}}{\sqrt{2}} = 100 \text{ V}$$
$$f = \frac{100\pi}{2\pi} = 50 \text{ Hz}$$

Final Answer

$V_{rms} = 100$ V, $f = 50$ Hz

Common Mistake: Do not confuse the coefficient $100\sqrt{2}$ with the rms value directly – it is the peak value; dividing by $\sqrt{2}$ gives rms.

Question 8

The current and voltage in a circuit are given by $I = 3.5 \sin(628t + 30^\circ)$ A, $V = 28 \sin(628t - 30^\circ)$ V. Determine (i) peak value, (ii) root-mean-square value, (iii) time period, and (iv) phase difference between voltage and current.

★ Likely Exam Question

Given: $I = 3.5 \sin(628t + 30^\circ)$ A, $V = 28 \sin(628t - 30^\circ)$ V

Formula: $I_{rms} = \frac{I_0}{\sqrt{2}}$, $T = \frac{2\pi}{\omega}$, $\Delta\phi = \phi_I - \phi_V$

Solution:

(i) Peak value of current: $I_0 = 3.5$ A.

(ii) rms value:

$$I_{rms} = \frac{3.5}{\sqrt{2}} = 2.47 \text{ A}$$

(iii) Time period: here $\omega = 628$ rad/s,

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{628} = 0.01 \text{ s}$$

(iv) Phase difference: current phase = $+30^\circ$, voltage phase = -30° ,

$$\Delta\phi = 30^\circ - (-30^\circ) = 60^\circ$$

Final Answer

(i) $I_0 = 3.5$ A; (ii) $I_{rms} = 2.47$ A; (iii) $T = 0.01$ s; (iv) $\Delta\phi = 60^\circ$ (current leads voltage)

Common Mistake: Phase difference must be found by subtracting the two phase angles algebraically, keeping track of signs.

II. AC Circuit with either of R, L or C

Question 9

What is the self-inductance of a coil if its inductive reactance is 12.56Ω at a frequency of 200 Hz?

Given: $X_L = 12.56 \Omega$, $f = 200 \text{ Hz}$

Formula: $X_L = 2\pi fL$

Solution:

$$L = \frac{X_L}{2\pi f} = \frac{12.56}{2\pi \times 200} = \frac{12.56}{1256.6} = 0.01 \text{ H}$$

Final Answer

$L = 0.01 \text{ H}$

Question 10

At what frequency will a 2 H inductor have a reactance of 4000Ω ?

Given: $L = 2 \text{ H}$, $X_L = 4000 \Omega$

Formula: $X_L = 2\pi fL$

Solution:

$$f = \frac{X_L}{2\pi L} = \frac{4000}{2\pi \times 2} = \frac{4000}{4\pi} = \frac{1000}{\pi} \text{ Hz}$$

Final Answer

$f = \frac{1000}{\pi} \text{ Hz} \approx 318.3 \text{ Hz}$

Question 11

A capacitor of 50 pF is connected to an AC source of frequency 1 kHz. Calculate its reactance.

Given: $C = 50 \text{ pF} = 50 \times 10^{-12} \text{ F}$, $f = 1 \text{ kHz} = 1000 \text{ Hz}$

Formula: $X_C = \frac{1}{2\pi fC}$

Solution:

$$X_C = \frac{1}{2\pi \times 1000 \times 50 \times 10^{-12}} = \frac{1}{2\pi \times 5 \times 10^{-8}} = \frac{10^7}{\pi} \Omega$$

Final Answer

$$X_C = \frac{10^7}{\pi} \Omega \approx 3.18 \times 10^6 \Omega$$

Question 12

A pure inductor of 0.5 H carries an AC of 0.2 A and frequency 50 Hz. What will be the voltage across the ends of the inductor and the phase difference between the voltage and the current?

Given: $L = 0.5 \text{ H}$, $I = 0.2 \text{ A}$, $f = 50 \text{ Hz}$

Formula: $X_L = 2\pi fL$, $V = IX_L$

Solution:

$$X_L = 2\pi \times 50 \times 0.5 = 157.1 \Omega$$

$$V = IX_L = 0.2 \times 157.1 = 31.4 \text{ V}$$

In a pure inductor the current lags behind the voltage by 90° .

Final Answer

$V = 31.4 \text{ V}$, phase difference = 90° (current lags voltage)

Question 13

A pure inductance of 1.0 H is connected across a 110 V – 70 Hz source. Find the reactance, current, and peak value of current.

Given: $L = 1.0 \text{ H}$, $V = 110 \text{ V (rms)}$, $f = 70 \text{ Hz}$

Formula: $X_L = 2\pi fL$, $I = \frac{V}{X_L}$, $I_0 = I\sqrt{2}$

Solution:

$$X_L = 2\pi \times 70 \times 1.0 = 439.8 \Omega \approx 440 \Omega$$

$$I = \frac{110}{440} = 0.25 \text{ A}$$

$$I_0 = 0.25 \times \sqrt{2} = 0.354 \text{ A}$$

Final Answer

$$X_L = 440 \Omega, \quad I = 0.25 \text{ A (rms)}, \quad I_0 = 0.35 \text{ A}$$

Question 14

A $2.0 \mu\text{F}$ capacitor is connected to a 220 V , 50 Hz AC source. Calculate the reactance of the capacitor, the current flowing in the circuit, and mention the phase relationship between the current and the voltage.

Given: $C = 2.0 \mu\text{F} = 2 \times 10^{-6} \text{ F}$, $V = 220 \text{ V}$, $f = 50 \text{ Hz}$

Formula: $X_C = \frac{1}{2\pi fC}$, $I = \frac{V}{X_C}$

Solution:

$$X_C = \frac{1}{2\pi \times 50 \times 2 \times 10^{-6}} = \frac{1}{6.283 \times 10^{-4}} = 1592.3 \Omega$$

$$I = \frac{220}{1592.3} = 0.138 \text{ A}$$

In a pure capacitor, the current leads the voltage by 90° .

Final Answer

$$X_C = 1592.3 \Omega, \quad I = 0.138 \text{ A}, \quad \text{current leads voltage by } 90^\circ$$

Common Mistake: A common exam trap is writing "voltage leads current" for a capacitor – remember: *C*apacitor, *C*urrent leads.

Question 15

Find the frequency at which the capacitive reactance of a $1.0 \mu\text{F}$ capacitor equals the inductive reactance of a 1.0 H coil.

Given: $C = 1.0 \mu\text{F} = 10^{-6} \text{ F}$, $L = 1.0 \text{ H}$

Formula: $X_L = X_C \Rightarrow f = \frac{1}{2\pi\sqrt{LC}}$

Solution:

$$\sqrt{LC} = \sqrt{1 \times 10^{-6}} = 10^{-3}$$

$$f = \frac{1}{2\pi \times 10^{-3}} = 159.24 \text{ Hz}$$

Final Answer

$$f = 159.24 \text{ Hz}$$

Question 16

A coil has an inductance of 1 H. At what frequency will it have a reactance of $3140\ \Omega$? What capacitance will have the same reactance at that frequency?

Given: $L = 1\ \text{H}$, $X_L = 3140\ \Omega$

Formula: $X_L = 2\pi fL$, $X_C = \frac{1}{2\pi fC}$

Solution:

Frequency:

$$f = \frac{X_L}{2\pi L} = \frac{3140}{2\pi} = 500\ \text{Hz}$$

Capacitance for the same reactance at this frequency:

$$C = \frac{1}{2\pi fX_C} = \frac{1}{2\pi \times 500 \times 3140} = 1.01 \times 10^{-7}\ \text{F}$$

Final Answer

$$f = 500\ \text{Hz}, \quad C \approx 1 \times 10^{-7}\ \text{F}$$

III. AC Circuit containing R and L

Question 17

A 100 V – 50 Hz AC source is connected to a series combination of an inductance of 100 mH and a resistance of 25 Ω . Calculate the magnitude and phase of current.

★ Likely Exam Question

Given: $V = 100$ V, $f = 50$ Hz, $L = 100$ mH = 0.1 H, $R = 25$ Ω

Formula: $X_L = 2\pi fL$, $Z = \sqrt{R^2 + X_L^2}$, $I = \frac{V}{Z}$, $\tan \phi = \frac{X_L}{R}$

Solution:

$$X_L = 2\pi \times 50 \times 0.1 = 31.4 \Omega$$

$$Z = \sqrt{25^2 + 31.4^2} = \sqrt{625 + 986} = \sqrt{1611} = 40.1 \Omega$$

$$I = \frac{100}{40.1} = 2.5 \text{ A}$$

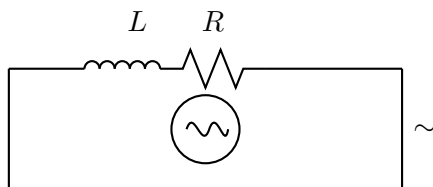
$$\tan \phi = \frac{31.4}{25} = 1.256 \Rightarrow \phi = 51.5^\circ$$

Final Answer

$I = 2.5$ A, $\phi = 51.5^\circ$ (current lags behind voltage)

Question 18

A current of 2.0 A is flowing in the given L-R circuit. Find the voltage across the AC source and the impedance of the circuit. ($V_L = 240$ V, $V_R = 100$ V)



Series L-R circuit ($V_L = 240$ V, $V_R = 100$ V) fed from an AC source

Given: $I = 2.0$ A, $V_L = 240$ V, $V_R = 100$ V

Formula: $V = \sqrt{V_L^2 + V_R^2}, \quad Z = \frac{V}{I}$

Solution:

$$V = \sqrt{240^2 + 100^2} = \sqrt{57600 + 10000} = \sqrt{67600} = 260 \text{ V}$$

$$Z = \frac{260}{2.0} = 130 \Omega$$

Final Answer

$$V = 260 \text{ V}, \quad Z = 130 \Omega$$

Common Mistake: Since V_L and V_R are 90° out of phase, they must be added as $\sqrt{V_L^2 + V_R^2}$, never arithmetically.

Question 19

An alternating emf of 110 V is applied to a circuit containing a resistance of 40Ω and an inductance L in series. The current lags behind the emf by an angle $\tan^{-1}(3/4)$. Find the inductive reactance, impedance of the circuit and current in the circuit. If $L = 0.1 \text{ H}$, find the (angular) frequency of applied emf.

★ Likely Exam Question

Given: $V = 110 \text{ V}, \quad R = 40 \Omega, \quad \tan \phi = 3/4, \quad L = 0.1 \text{ H}$

Formula: $\tan \phi = \frac{X_L}{R}, \quad Z = \sqrt{R^2 + X_L^2}, \quad I = \frac{V}{Z}, \quad X_L = \omega L$

Solution:

$$X_L = R \tan \phi = 40 \times \frac{3}{4} = 30 \Omega$$

$$Z = \sqrt{40^2 + 30^2} = \sqrt{1600 + 900} = \sqrt{2500} = 50 \Omega$$

$$I = \frac{110}{50} = 2.2 \text{ A}$$

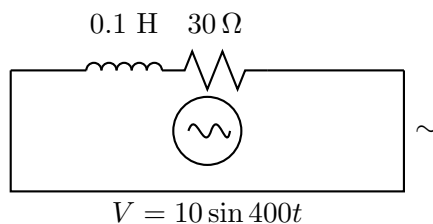
$$\omega = \frac{X_L}{L} = \frac{30}{0.1} = 300 \text{ rad/s}$$

Final Answer

$$X_L = 30 \Omega, \quad Z = 50 \Omega, \quad I = 2.2 \text{ A}, \quad \omega = 300 \text{ rad s}^{-1}$$

Question 20

For the given circuit ($L = 0.1 \text{ H}, R = 30 \Omega, V = 10 \sin 400t$), find (i) the reactance, (ii) the impedance, (iii) the peak value of current and (iv) the phase difference between voltage and current. ($\tan^{-1} 1.33 = 53^\circ$)



Series circuit with $L = 0.1 \text{ H}$ and $R = 30 \Omega$, source $V = 10 \sin 400t$

Given: $L = 0.1 \text{ H}$, $R = 30 \Omega$, $V = 10 \sin 400t$ (so $V_0 = 10 \text{ V}$, $\omega = 400 \text{ rad/s}$)

Formula: $X_L = \omega L$, $Z = \sqrt{R^2 + X_L^2}$, $I_0 = \frac{V_0}{Z}$, $\tan \phi = \frac{X_L}{R}$

Solution:

(i) $X_L = \omega L = 400 \times 0.1 = 40 \Omega$

(ii) $Z = \sqrt{30^2 + 40^2} = \sqrt{900 + 1600} = \sqrt{2500} = 50 \Omega$

(iii) $I_0 = \frac{10}{50} = 0.2 \text{ A}$

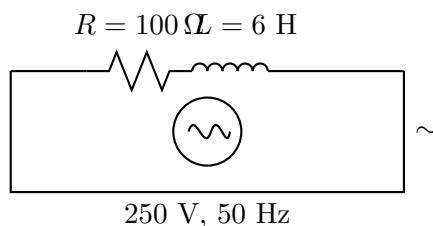
(iv) $\tan \phi = \frac{40}{30} = 1.33 \Rightarrow \phi = 53^\circ$

Final Answer

(i) $X_L = 40 \Omega$; (ii) $Z = 50 \Omega$; (iii) $I_0 = 0.2 \text{ A}$; (iv) $\phi = 53^\circ$ (current lags)

Question 21

An AC circuit has a resistance and an inductance connected in series, as shown ($R = 100 \Omega$, $L = 6 \text{ H}$, 250 V , 50 Hz). Calculate the current and the power factor in the circuit.



Series R-L circuit ($R = 100 \Omega$, $L = 6 \text{ H}$) fed by 250 V , 50 Hz AC mains

Given: $R = 100 \Omega$, $L = 6 \text{ H}$, $V = 250 \text{ V}$, $f = 50 \text{ Hz}$

Formula: $X_L = 2\pi f L$, $Z = \sqrt{R^2 + X_L^2}$, $I = \frac{V}{Z}$, $\cos \phi = \frac{R}{Z}$

Solution:

$$X_L = 2\pi \times 50 \times 6 = 1885 \Omega$$

$$Z = \sqrt{100^2 + 1885^2} = \sqrt{10000 + 3553225} = \sqrt{3563225} = 1887.6 \Omega$$

$$I = \frac{250}{1887.6} = 0.1325 \text{ A}$$

$$\cos \phi = \frac{100}{1887.6} = 0.053$$

Final Answer

$$I = 0.1325 \text{ A}, \quad \cos \phi = 0.053$$

Question 22

A coil has a current of 1.0 A and a power of 100 W from an AC source of 110 V and 50 Hz. Find the resistance and the inductance of the coil.

★ Likely Exam Question

$$\text{Given: } I = 1.0 \text{ A}, \quad P = 100 \text{ W}, \quad V = 110 \text{ V}, \quad f = 50 \text{ Hz}$$

$$\text{Formula: } P = I^2 R, \quad Z = \frac{V}{I}, \quad X_L = \sqrt{Z^2 - R^2}, \quad L = \frac{X_L}{2\pi f}$$

Solution:

$$R = \frac{P}{I^2} = \frac{100}{1^2} = 100 \Omega$$

$$Z = \frac{V}{I} = \frac{110}{1} = 110 \Omega$$

$$X_L = \sqrt{110^2 - 100^2} = \sqrt{12100 - 10000} = \sqrt{2100} = 45.8 \Omega$$

$$L = \frac{45.8}{2\pi \times 50} = \frac{45.8}{314.16} = 0.146 \text{ H}$$

Final Answer

$$R = 100 \Omega, \quad L = 0.146 \text{ H}$$

Common Mistake: Power in an R - L circuit depends only on R (i.e. $P = I^2 R$); do not use Z in place of R in the power formula.

Question 23

An emf $V = 200 \sin 377t$ volt is applied across an inductance L having a resistance of 1.0Ω . The maximum current is found to be 10 A. Find the value of L .

$$\text{Given: } V_0 = 200 \text{ V}, \quad \omega = 377 \text{ rad/s}, \quad R = 1.0 \Omega, \quad I_0 = 10 \text{ A}$$

Formula: $Z = \frac{V_0}{I_0}$, $X_L = \sqrt{Z^2 - R^2}$, $L = \frac{X_L}{\omega}$

Solution:

$$Z = \frac{200}{10} = 20 \Omega$$

$$X_L = \sqrt{20^2 - 1^2} = \sqrt{400 - 1} = \sqrt{399} = 19.97 \Omega$$

$$L = \frac{19.97}{377} = 0.053 \text{ H} = 53 \text{ mH}$$

Final Answer

$$L = 53 \text{ mH}$$

Question 24

An electric lamp which runs at 80 V DC, consumes 10 A current. The lamp is connected to a 100 V – 50 Hz AC source. Compute the inductance of the choke required (so the lamp operates normally).

Given: DC: $V = 80 \text{ V}$, $I = 10 \text{ A}$; AC: $V = 100 \text{ V}$, $f = 50 \text{ Hz}$, same $I = 10 \text{ A}$

Formula: $R = \frac{V_{dc}}{I}$, $Z = \frac{V_{ac}}{I}$, $X_L = \sqrt{Z^2 - R^2}$, $L = \frac{X_L}{2\pi f}$

Solution:

Resistance of lamp: $R = \frac{80}{10} = 8 \Omega$

Impedance required on AC: $Z = \frac{100}{10} = 10 \Omega$

$$X_L = \sqrt{10^2 - 8^2} = \sqrt{100 - 64} = \sqrt{36} = 6 \Omega$$

$$L = \frac{6}{2\pi \times 50} = \frac{6}{314.16} = 0.02 \text{ H}$$

Final Answer

$$L = 0.02 \text{ H}$$

Common Mistake: The choke (an ideal inductor) does not dissipate power; only R of the lamp accounts for power, but L must be added in quadrature to limit current.

Question 25

An inductor-resistor (L-R) series circuit is connected to an AC source of 220 V, 50 Hz. The resistor value is 11 ohms, and the power factor is $1/\sqrt{2}$. Calculate the value of inductance in the circuit.

Given: $R = 11 \Omega$, $\cos \phi = \frac{1}{\sqrt{2}}$, $f = 50 \text{ Hz}$

Formula: $\cos \phi = \frac{1}{\sqrt{2}} \Rightarrow \phi = 45^\circ$, $\tan \phi = \frac{X_L}{R}$, $L = \frac{X_L}{2\pi f}$

Solution:

$$\phi = 45^\circ \Rightarrow \tan \phi = 1 = \frac{X_L}{R} \Rightarrow X_L = R = 11 \Omega$$

$$L = \frac{11}{2\pi \times 50} = \frac{11}{314.16} = 0.035 \text{ H}$$

Final Answer

$$L = 0.035 \text{ H}$$

Question 26

A non-inductive resistance of 12Ω and an inductance of 0.05 H of negligible resistance are connected in series in an AC circuit having frequency 50 Hz and voltage 130 V . Determine impedance and current in the circuit.

Given: $R = 12 \Omega$, $L = 0.05 \text{ H}$, $f = 50 \text{ Hz}$, $V = 130 \text{ V}$

Formula: $X_L = 2\pi fL$, $Z = \sqrt{R^2 + X_L^2}$, $I = \frac{V}{Z}$

Solution:

$$X_L = 2\pi \times 50 \times 0.05 = 15.71 \Omega$$

$$Z = \sqrt{12^2 + 15.71^2} = \sqrt{144 + 246.8} = \sqrt{390.8} = 19.76 \Omega$$

$$I = \frac{130}{19.76} = 6.58 \text{ A}$$

Final Answer

$$Z = 19.76 \Omega, \quad I = 6.58 \text{ A}$$

Question 27

A bulb of resistance 10Ω , connected to an inductor of inductance L , is in series with an AC source marked 100 V , 50 Hz . If the phase angle between the voltage and the current is $\pi/4$ radian, find the value of L .

Given: $R = 10 \Omega$, $V = 100 \text{ V}$, $f = 50 \text{ Hz}$, $\phi = \pi/4 = 45^\circ$

Formula: $\tan \phi = \frac{X_L}{R}$, $L = \frac{X_L}{2\pi f}$

Solution:

$$X_L = R \tan 45^\circ = 10 \times 1 = 10 \Omega$$

$$L = \frac{10}{2\pi \times 50} = \frac{10}{314.16} = \frac{1}{10\pi} \text{ H} \approx 0.0318 \text{ H}$$

Final Answer

$$L = \frac{1}{10\pi} \text{ H}$$

Question 28

A current of 1.1 A flows through a coil when connected to a 110 V DC source. When 110 V-50 Hz AC is applied to the same coil, only 0.5 A current flows. Calculate the resistance, impedance and inductance of the coil.

★ **Likely Exam Question**

Given: DC: $V = 110 \text{ V}$, $I = 1.1 \text{ A}$; AC: $V = 110 \text{ V}$, $f = 50 \text{ Hz}$, $I = 0.5 \text{ A}$

Formula: $R = \frac{V_{dc}}{I_{dc}}$, $Z = \frac{V_{ac}}{I_{ac}}$, $X_L = \sqrt{Z^2 - R^2}$, $L = \frac{X_L}{2\pi f}$

Solution:

$$R = \frac{110}{1.1} = 100 \Omega$$

$$Z = \frac{110}{0.5} = 220 \Omega$$

$$X_L = \sqrt{220^2 - 100^2} = \sqrt{48400 - 10000} = \sqrt{38400} = 196.0 \Omega$$

$$L = \frac{196.0}{2\pi \times 50} = \frac{196.0}{314.16} = 0.624 \text{ H}$$

Final Answer

$$R = 100 \Omega, \quad Z = 220 \Omega, \quad L = 0.624 \text{ H}$$

Common Mistake: On DC, an inductor offers no opposition ($X_L = 0$), so the DC test gives R directly; do not use the AC current value to find R .

Question 29

An AC circuit consists of a 220 V, 50 Hz supply connected across a 100Ω resistor. What inductance should be introduced in the circuit in series with the resistance so that the current is reduced to half?

Given: $V = 220 \text{ V}$, $f = 50 \text{ Hz}$, $R = 100 \Omega$

Formula: $I_1 = \frac{V}{R}, \quad I_2 = \frac{I_1}{2}, \quad Z = \frac{V}{I_2}, \quad X_L = \sqrt{Z^2 - R^2}, \quad L = \frac{X_L}{2\pi f}$

Solution:

Original current: $I_1 = \frac{220}{100} = 2.2$ A. Required current: $I_2 = 1.1$ A.

$$Z = \frac{220}{1.1} = 200 \, \Omega$$

$$X_L = \sqrt{200^2 - 100^2} = \sqrt{40000 - 10000} = \sqrt{30000} = 173.2 \, \Omega$$

$$L = \frac{173.2}{2\pi \times 50} = \frac{173.2}{314.16} = 0.551 \, \text{H}$$

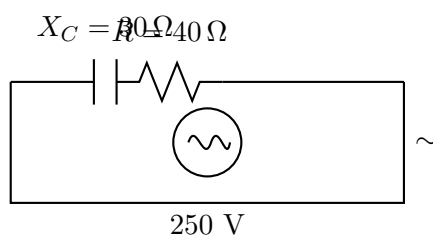
Final Answer

$$L = 0.55 \, \text{H}$$

IV. AC Circuit containing R and C

Question 30

In the given AC circuit ($X_C = 30\ \Omega$, $R = 40\ \Omega$, $V = 250\ \text{V}$), find (i) the potential difference across the resistor, and (ii) across the capacitor.



Series circuit with $X_C = 30\ \Omega$ and $R = 40\ \Omega$, source $250\ \text{V}$

Given: $X_C = 30\ \Omega$, $R = 40\ \Omega$, $V = 250\ \text{V}$

Formula: $Z = \sqrt{R^2 + X_C^2}$, $I = \frac{V}{Z}$, $V_R = IR$, $V_C = IX_C$

Solution:

$$Z = \sqrt{40^2 + 30^2} = \sqrt{1600 + 900} = \sqrt{2500} = 50\ \Omega$$

$$I = \frac{250}{50} = 5\ \text{A}$$

$$V_R = 5 \times 40 = 200\ \text{V}, \quad V_C = 5 \times 30 = 150\ \text{V}$$

Final Answer

(i) $V_R = 200\ \text{V}$; (ii) $V_C = 150\ \text{V}$

Question 31

A $5\ \text{V}$ AC source is connected to a C-R circuit. The PD across C is $3\ \text{V}$. Find the PD across R .

Given: $V = 5\ \text{V}$, $V_C = 3\ \text{V}$

Formula: $V = \sqrt{V_R^2 + V_C^2}$

Solution:

$$V_R = \sqrt{V^2 - V_C^2} = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4\ \text{V}$$

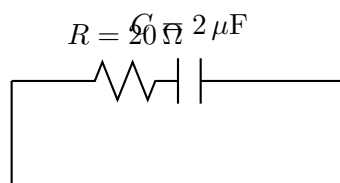
Final Answer

$$V_R = 4 \text{ V}$$

Common Mistake: V_R and V_C are 90° out of phase, so they combine as a right-angle triangle ($V^2 = V_R^2 + V_C^2$), never by simple addition.

Question 32

Find the impedance of the given circuit ($R = 20 \Omega$, $C = 2 \mu\text{F}$) for (i) direct current, and (ii) alternating current of frequency $(10/\pi) \text{ kHz}$.



Series R-C circuit ($R = 20 \Omega$, $C = 2 \mu\text{F}$)

Given: $R = 20 \Omega$, $C = 2 \mu\text{F} = 2 \times 10^{-6} \text{ F}$, $f = \frac{10}{\pi} \text{ kHz} = \frac{10^4}{\pi} \text{ Hz}$

Formula: $X_C = \frac{1}{2\pi fC}$, $Z = \sqrt{R^2 + X_C^2}$

Solution:

(i) For DC, $f = 0$, so $X_C = 1/(2\pi fC) \rightarrow \infty$. A capacitor blocks DC completely, hence impedance is infinite.

(ii) For AC:

$$2\pi f = 2\pi \times \frac{10^4}{\pi} = 2 \times 10^4 \text{ rad/s}$$

$$X_C = \frac{1}{2 \times 10^4 \times 2 \times 10^{-6}} = \frac{1}{0.04} = 25 \Omega$$

$$Z = \sqrt{20^2 + 25^2} = \sqrt{400 + 625} = \sqrt{1025} = 32.0 \Omega$$

Final Answer

(i) $Z = \infty$ (capacitor blocks DC); (ii) $Z = 32 \Omega$

Common Mistake: A capacitor offers infinite reactance to DC ($f = 0$), so no steady current can ever flow through a capacitor branch on DC.

Question 33

A capacitor in series with a resistance of 30Ω is connected to AC mains. The reactance of the capacitor is 40Ω . Calculate the phase difference between the current and the supply voltage.

Given: $R = 30\ \Omega$, $X_C = 40\ \Omega$

Formula: $\tan \phi = \frac{X_C}{R}$

Solution:

$$\tan \phi = \frac{40}{30} = \frac{4}{3} \Rightarrow \phi = \tan^{-1} \left(\frac{4}{3} \right)$$

Final Answer

$\phi = \tan^{-1}(4/3)$ (current leads the voltage)

Question 34

An alternating emf of $100\text{ V} - 50\text{ Hz}$ is applied across a $10\ \mu\text{F}$ capacitor and $100\ \Omega$ resistor in series. Calculate the reactance of the capacitor, the current flowing, and the average power supplied.

Given: $V = 100\text{ V}$, $f = 50\text{ Hz}$, $C = 10\ \mu\text{F} = 10 \times 10^{-6}\text{ F}$, $R = 100\ \Omega$

Formula: $X_C = \frac{1}{2\pi fC}$, $Z = \sqrt{R^2 + X_C^2}$, $I = \frac{V}{Z}$, $P = I^2 R$

Solution:

$$X_C = \frac{1}{2\pi \times 50 \times 10 \times 10^{-6}} = \frac{1}{3.1416 \times 10^{-3}} = 318.3\ \Omega$$

$$Z = \sqrt{100^2 + 318.3^2} = \sqrt{10000 + 101314} = \sqrt{111314} = 333.6\ \Omega$$

$$I = \frac{100}{333.6} = 0.30\text{ A}$$

$$P = I^2 R = (0.30)^2 \times 100 = 9.0\text{ W}$$

Final Answer

$X_C = 318\ \Omega$, $I = 0.30\text{ A}$, $P = 9.0\text{ W}$

Common Mistake: Average power depends only on R (as $I^2 R$); a capacitor consumes no average power since $\cos \phi$ contribution from it cancels over a full cycle.

V. L-C-R Circuit

Question 35

In an AC circuit with all components connected in series, the emf and the current are given by $V = 200 \sin \left(314t + \frac{\pi}{6} \right)$ volt and $I = 5 \sin 314t$ ampere. Obtain (i) the peak values of emf and current, (ii) the frequency of the AC source, (iii) the phase difference between V and I , and (iv) the impedance of the circuit. (ISC 2000)

★ Likely Exam Question

Given: $V = 200 \sin \left(314t + \frac{\pi}{6} \right)$, $I = 5 \sin 314t$

Formula: $f = \frac{\omega}{2\pi}$, $Z = \frac{V_0}{I_0}$

Solution:

(i) Peak emf $V_0 = 200$ V; peak current $I_0 = 5$ A.

(ii) $f = \frac{\omega}{2\pi} = \frac{314}{2\pi} = 50$ Hz

(iii) Voltage phase = $\pi/6$, current phase = 0, so phase difference = $\pi/6$ (emf leads current).

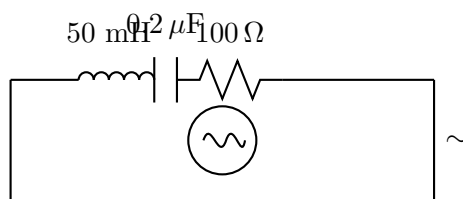
(iv) $Z = \frac{V_0}{I_0} = \frac{200}{5} = 40 \Omega$

Final Answer

(i) $V_0 = 200$ V, $I_0 = 5$ A; (ii) $f = 50$ Hz; (iii) $\pi/6$; (iv) $Z = 40 \Omega$

Question 36

In the given circuit ($L = 50$ mH, $C = 0.2 \mu\text{F}$, $R = 100 \Omega$, $V = 30 \sin 1000t$), calculate the inductive reactance and the capacitive reactance. What should be the source frequency so that the two reactances become equal in magnitude? What would then be the impedance of the circuit?



Series L-C-R circuit ($L = 50 \text{ mH}$, $C = 0.2 \mu\text{F}$, $R = 100 \Omega$), source $V = 30 \sin 1000t$

Given: $L = 50 \text{ mH} = 0.05 \text{ H}$, $C = 0.2 \mu\text{F} = 0.2 \times 10^{-6} \text{ F}$, $R = 100 \Omega$, $\omega = 1000 \text{ rad/s}$

Formula: $X_L = \omega L$, $X_C = \frac{1}{\omega C}$, $X_L = X_C \Rightarrow \omega = \frac{1}{\sqrt{LC}}$, $Z_{res} = R$

Solution:

At $\omega = 1000 \text{ rad/s}$:

$$X_L = 1000 \times 0.05 = 50 \Omega$$

$$X_C = \frac{1}{1000 \times 0.2 \times 10^{-6}} = \frac{1}{2 \times 10^{-4}} = 5000 \Omega$$

For $X_L = X_C$:

$$LC = 0.05 \times 0.2 \times 10^{-6} = 10^{-8}, \quad \omega = \frac{1}{\sqrt{LC}} = \frac{1}{10^{-4}} = 10^4 \text{ rad/s}$$

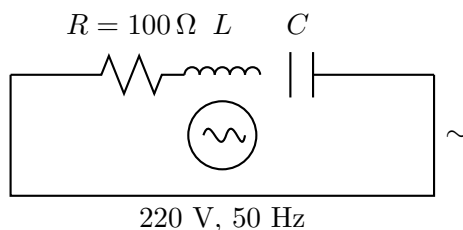
At this frequency, reactances cancel, so $Z = R = 100 \Omega$.

Final Answer

$X_L = 50 \Omega$, $X_C = 5000 \Omega$ (at $\omega = 1000$); for equal reactance $\omega = 10^4 \text{ rad s}^{-1}$, and then $Z = R = 100 \Omega$

Question 37

In the given L-C-R series circuit fed by a 220 V – 50 Hz AC mains, find (i) the current in the circuit, (ii) the value of inductor L , (iii) the value of capacitor C , and (iv) the value of C (for the same L) required to produce resonance. The voltmeter readings across R ($= 100 \Omega$), L and C are 65 V, 204 V and 415 V respectively.



Series L-C-R circuit fed by 220 V, 50 Hz mains; voltmeters
read $V_R = 65 \text{ V}$, $V_L = 204 \text{ V}$, $V_C = 415 \text{ V}$

Given: $R = 100 \Omega$, $f = 50 \text{ Hz}$, $V_R = 65 \text{ V}$, $V_L = 204 \text{ V}$, $V_C = 415 \text{ V}$

Formula: $I = \frac{V_R}{R}$, $X_L = \frac{V_L}{I}$, $L = \frac{X_L}{2\pi f}$, $X_C = \frac{V_C}{I}$, $C = \frac{1}{2\pi f X_C}$

Solution:

$$(i) I = \frac{65}{100} = 0.65 \text{ A}$$

$$(ii) X_L = \frac{204}{0.65} = 313.8 \Omega \Rightarrow L = \frac{313.8}{2\pi \times 50} = \frac{313.8}{314.16} \approx 1.0 \text{ H}$$

$$(iii) X_C = \frac{415}{0.65} = 638.5 \Omega \Rightarrow C = \frac{1}{2\pi \times 50 \times 638.5} = \frac{1}{200600} \approx 5 \mu\text{F}$$

(iv) For resonance with the same L : need $X_C = X_L = 313.8 \Omega$ at 50 Hz,

$$C = \frac{1}{2\pi \times 50 \times 313.8} = \frac{1}{98650} \approx 10 \mu\text{F}$$

Final Answer

(i) $I = 0.65 \text{ A}$; (ii) $L \approx 1.0 \text{ H}$; (iii) $C \approx 5 \mu\text{F}$; (iv) $C_{res} \approx 10 \mu\text{F}$

Common Mistake: Current is the same everywhere in a series circuit – use $I = V_R/R$ (not V/Z of total emf) to start, since R 's voltmeter directly gives I .

Question 38

A resistor of 50Ω , an inductor of $(20/\pi) \text{ H}$ and a capacitor of $(5/\pi) \mu\text{F}$ are connected in series to a voltage supply of $230 \text{ V} - 50 \text{ Hz}$. Find the impedance of the circuit.

Given: $R = 50 \Omega$, $L = \frac{20}{\pi} \text{ H}$, $C = \frac{5}{\pi} \mu\text{F}$, $f = 50 \text{ Hz}$

Formula: $X_L = 2\pi f L$, $X_C = \frac{1}{2\pi f C}$, $Z = \sqrt{R^2 + (X_L - X_C)^2}$

Solution:

$$X_L = 2\pi \times 50 \times \frac{20}{\pi} = 100 \times 20 = 2000 \Omega$$

$$X_C = \frac{1}{2\pi \times 50 \times \frac{5}{\pi} \times 10^{-6}} = \frac{1}{100 \times 5 \times 10^{-6}} = \frac{1}{5 \times 10^{-4}} = 2000 \Omega$$

Since $X_L = X_C$, they cancel: $Z = R = 50 \Omega$.

Final Answer

$$Z = 50 \Omega$$

Common Mistake: Whenever L and C values are given with a π in them, check whether X_L and X_C are designed to cancel exactly – a common exam trick.

Question 39

A $50 \mu\text{F}$ capacitor, a 0.05 H inductor and a 48Ω resistor are in series with an AC source of emf $V = 310 \sin 314t$. Calculate the reactance of the circuit and its character. What is the phase angle between the current and the applied emf?

Given: $C = 50 \mu\text{F}$, $L = 0.05 \text{ H}$, $R = 48 \Omega$, $\omega = 314 \text{ rad/s}$

Formula: $X_L = \omega L$, $X_C = \frac{1}{\omega C}$, $X = X_C - X_L$, $\tan \phi = \frac{X}{R}$

Solution:

$$X_L = 314 \times 0.05 = 15.7 \Omega$$

$$X_C = \frac{1}{314 \times 50 \times 10^{-6}} = \frac{1}{0.0157} = 63.7 \Omega$$

Since $X_C > X_L$, the circuit is capacitive:

$$X = X_C - X_L = 63.7 - 15.7 = 48 \Omega \text{ (capacitive)}$$

$$\tan \phi = \frac{48}{48} = 1 \Rightarrow \phi = 45^\circ$$

Final Answer

$X = 48 \Omega$ (capacitive), $\phi = 45^\circ$ (current leads voltage)

Question 40

A capacitor, a resistor of 5Ω and an inductor of 50 mH are in series with an AC source marked $100 \text{ V}, 50 \text{ Hz}$. It is found that voltage is in phase with the current. Calculate the capacitance of the capacitor and the impedance of the circuit. (Hint: $\omega L = 1/\omega C$)

Given: $R = 5 \Omega$, $L = 50 \text{ mH} = 0.05 \text{ H}$, $f = 50 \text{ Hz}$, V, I in phase (resonance)

Formula: $\omega L = \frac{1}{\omega C} \Rightarrow C = \frac{1}{\omega^2 L}$, $Z_{res} = R$

Solution:

$$\omega = 2\pi f = 2\pi \times 50 = 314.16 \text{ rad/s}$$

$$C = \frac{1}{\omega^2 L} = \frac{1}{(314.16)^2 \times 0.05} = \frac{1}{4934.8} = 2.026 \times 10^{-4} \text{ F} = 202.6 \mu\text{F}$$

Since V, I are in phase, this is resonance, so:

$$Z = R = 5 \Omega$$

Final Answer

$$C \approx 202.8 \mu\text{F}, \quad Z = 5 \Omega$$

Common Mistake: "Voltage in phase with current" is the defining condition of resonance ($X_L = X_C$); do not try to compute X_L and X_C separately as unequal quantities.

Question 41

In an L-C-R AC circuit, the reactance of L and C are 160Ω and 130Ω respectively, and R is 40Ω . Find the impedance of the circuit.

$$\text{Given: } X_L = 160 \Omega, \quad X_C = 130 \Omega, \quad R = 40 \Omega$$

$$\text{Formula: } Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Solution:

$$Z = \sqrt{40^2 + (160 - 130)^2} = \sqrt{1600 + 900} = \sqrt{2500} = 50 \Omega$$

Final Answer

$$Z = 50 \Omega$$

Question 42

A 40Ω resistor, a 3 mH inductor and a $2 \mu\text{F}$ capacitor are connected in series to a $110 \text{ V} - 5000 \text{ Hz}$ AC source. Find the current in the circuit.

$$\text{Given: } R = 40 \Omega, \quad L = 3 \text{ mH} = 0.003 \text{ H}, \quad C = 2 \mu\text{F}, \quad V = 110 \text{ V}, \quad f = 5000 \text{ Hz}$$

$$\text{Formula: } X_L = 2\pi fL, \quad X_C = \frac{1}{2\pi fC}, \quad Z = \sqrt{R^2 + (X_L - X_C)^2}, \quad I = \frac{V}{Z}$$

Solution:

$$\omega = 2\pi \times 5000 = 31416 \text{ rad/s}$$

$$X_L = 31416 \times 0.003 = 94.25 \Omega$$

$$X_C = \frac{1}{31416 \times 2 \times 10^{-6}} = \frac{1}{0.0628} = 15.92 \Omega$$

$$Z = \sqrt{40^2 + (94.25 - 15.92)^2} = \sqrt{1600 + 6135.6} = \sqrt{7735.6} = 88.0 \Omega$$

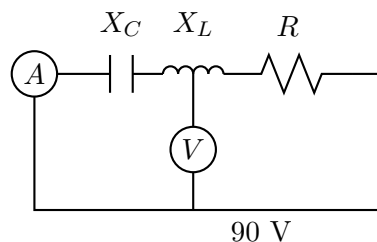
$$I = \frac{110}{88.0} = 1.25 \text{ A}$$

Final Answer

$$I = 1.25 \text{ A}$$

Question 43

What will be the readings in the voltmeter and the ammeter shown in the figure?
($X_C = 4 \Omega$, $X_L = 4 \Omega$, $R = 45 \Omega$, source = 90 V)



*Circuit with ammeter reading I and voltmeter across L - C
($X_C = 4 \Omega$, $X_L = 4 \Omega$, $R = 45 \Omega$, source 90 V)*

Given: $X_C = 4 \Omega$, $X_L = 4 \Omega$, $R = 45 \Omega$, $V = 90 \text{ V}$

Formula: $Z = \sqrt{R^2 + (X_L - X_C)^2}$, $I = \frac{V}{Z}$, $V_{(L \sim C)} = I(X_L - X_C)$

Solution:

Since $X_L = X_C = 4 \Omega$, they cancel exactly, so $Z = R = 45 \Omega$.

$$I = \frac{V}{Z} = \frac{90}{45} = 2 \text{ A}$$

Voltmeter across the L - C combination:

$$V_{(L \sim C)} = I(X_L - X_C) = 2 \times 0 = 0 \text{ V}$$

Final Answer

Ammeter reading $I = 2 \text{ A}$; Voltmeter reading across L - $C = 0 \text{ V}$

Common Mistake: Do not add X_L and X_C voltages directly; since they are 180° out of phase, equal reactances give a net PD of exactly zero.

Question 44

A $2 \mu\text{F}$ capacitor, 100Ω resistor and 8 H inductor are connected in series with an AC source. What should be the frequency of this source for which the current drawn in the circuit is maximum? If the peak value of emf of the source is 200 V , find for maximum current: (i) the inductive and capacitive reactance in the circuit, (ii) total impedance of the circuit, (iii) peak current in the circuit, (iv) phase relation between

voltages across inductor and resistor, and (v) phase difference between voltages across inductor and capacitor.

★ Likely Exam Question

Given: $C = 2 \mu\text{F}$, $R = 100 \Omega$, $L = 8 \text{ H}$, $V_0 = 200 \text{ V}$

Formula: $\omega_0 = \frac{1}{\sqrt{LC}}$ (resonance for max. current), $X_L = X_C = \omega_0 L$, $Z_{\min} = R$, $I_0 = \frac{V_0}{Z}$

Solution:

Maximum current occurs at resonance:

$$LC = 8 \times 2 \times 10^{-6} = 1.6 \times 10^{-5}, \quad \sqrt{LC} = 4 \times 10^{-3}$$

$$\omega_0 = \frac{1}{4 \times 10^{-3}} = 250 \text{ rad/s}$$

(i) $X_L = \omega_0 L = 250 \times 8 = 2000 \Omega$; and at resonance $X_C = X_L = 2000 \Omega$.

(ii) At resonance $Z = R = 100 \Omega$.

(iii) $I_0 = \frac{V_0}{Z} = \frac{200}{100} = 2 \text{ A}$.

(iv) V_L leads V_R by 90° (inductor voltage always leads the current, and V_R is in phase with current).

(v) V_L and V_C are exactly out of phase, so the phase difference between them is 180° .

Final Answer

$\omega_0 = 250 \text{ rad s}^{-1}$; (i) $X_L = X_C = 2000 \Omega$; (ii) $Z = 100 \Omega$; (iii) $I_0 = 2 \text{ A}$; (iv) 90° ; (v) 180°

VI. L-C Circuits and Oscillations

Question 45

In an L-C circuit connected to an AC source, the potential difference across L is 50 V and that across C is 30 V. Find the voltage of the source.

Given: $V_L = 50 \text{ V}$, $V_C = 30 \text{ V}$

Formula: $V = |V_L - V_C|$

Solution:

Since V_L and V_C are 180° out of phase in a pure L - C series circuit:

$$V = |V_L - V_C| = |50 - 30| = 20 \text{ V}$$

Final Answer

$$V = 20 \text{ V}$$

Common Mistake: In an L - C (no R) circuit, the two voltages subtract, unlike an R - L or R - C circuit where they combine at 90° .

Question 46

A $10 \mu\text{F}$ capacitor and a 1.0 H inductor are connected in series with an AC source of frequency 50 Hz . Find the impedance of the circuit. (Hint: $Z = \omega L - 1/\omega C$)

Given: $C = 10 \mu\text{F}$, $L = 1.0 \text{ H}$, $f = 50 \text{ Hz}$

Formula: $X_L = 2\pi fL$, $X_C = \frac{1}{2\pi fC}$, $Z = |X_L - X_C|$

Solution:

$$X_L = 2\pi \times 50 \times 1.0 = 314 \Omega$$

$$X_C = \frac{1}{2\pi \times 50 \times 10 \times 10^{-6}} = \frac{1}{0.00314} = 318.5 \Omega$$

$$Z = |X_L - X_C| = |314 - 318.5| = 4.5 \Omega \approx 4.47 \Omega$$

Final Answer

$$Z \approx 4.47 \Omega$$

Question 47

A 200 km long telegraph wire has a capacitance of $0.014 \mu\text{F}/\text{km}$. If it carries an AC of 5 kHz, what should be the inductance required to be connected in series so that the impedance is minimum? (Take $\pi = \sqrt{10}$)

Given: $C = 0.014 \mu\text{F}/\text{km} \times 200 \text{ km} = 2.8 \mu\text{F}$, $f = 5 \text{ kHz} = 5000 \text{ Hz}$, $\pi = \sqrt{10}$

Formula: $Z_{\min} \Rightarrow \omega L = \frac{1}{\omega C} \Rightarrow L = \frac{1}{\omega^2 C}$

Solution:

$$\omega = 2\pi f \Rightarrow \omega^2 = 4\pi^2 f^2 = 4 \times 10 \times (5000)^2 = 40 \times 25 \times 10^6 = 1 \times 10^9$$

$$L = \frac{1}{\omega^2 C} = \frac{1}{10^9 \times 2.8 \times 10^{-6}} = \frac{1}{2800} = 3.57 \times 10^{-4} \text{ H}$$

Final Answer

$$L \approx 0.357 \text{ mH}$$

Common Mistake: Total capacitance of a distributed cable is found by multiplying capacitance-per-km by the total length; do not use the per-km value directly.

Question 48

An AC source of frequency 50 Hz is connected to a 50 mH inductor and a bulb. The bulb glows with some brightness. Calculate the capacitance of the capacitor to be connected in series with the circuit so that the bulb glows with maximum brightness. (Hint: Z should be minimum, i.e. $\omega L = 1/\omega C$)

Given: $L = 50 \text{ mH} = 0.05 \text{ H}$, $f = 50 \text{ Hz}$

Formula: $\omega L = \frac{1}{\omega C} \Rightarrow C = \frac{1}{\omega^2 L}$

Solution:

$$\omega = 2\pi \times 50 = 314.16 \text{ rad/s}, \quad \omega^2 = 98696$$

$$C = \frac{1}{98696 \times 0.05} = \frac{1}{4934.8} = 2.026 \times 10^{-4} \text{ F}$$

Final Answer

$$C \approx 2.03 \times 10^{-4} \text{ F}$$

Common Mistake: Maximum brightness = maximum current = minimum impedance, i.e. resonance condition, not maximum reactance.

Question 49

Find the resonant angular frequency of a series L-C-R circuit with $L = 2.0 \text{ H}$, $C = 32 \mu\text{F}$ and $R = 10 \Omega$. What is the Q-value of this circuit?

Given: $L = 2.0 \text{ H}$, $C = 32 \mu\text{F} = 32 \times 10^{-6} \text{ F}$, $R = 10 \Omega$

Formula: $\omega_0 = \frac{1}{\sqrt{LC}}$, $Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$

Solution:

$$LC = 2.0 \times 32 \times 10^{-6} = 6.4 \times 10^{-5}, \quad \sqrt{LC} = 8 \times 10^{-3}$$

$$\omega_0 = \frac{1}{8 \times 10^{-3}} = 125 \text{ rad/s}$$

$$Q = \frac{\omega_0 L}{R} = \frac{125 \times 2.0}{10} = \frac{250}{10} = 25$$

Final Answer

$$\omega_0 = 125 \text{ rad s}^{-1}, \quad Q = 25$$

VII. Power Consumed in AC Circuits

Question 50

A sinusoidal voltage $V = 200 \sin 314t$ is applied to a 10Ω resistor. Find (i) rms voltage, (ii) rms current and (iii) power dissipated as heat.

Given: $V = 200 \sin 314t$, $R = 10 \Omega$

Formula: $V_{rms} = \frac{V_0}{\sqrt{2}}$, $I_{rms} = \frac{V_{rms}}{R}$, $P = I_{rms}^2 R$

Solution:

(i) $V_{rms} = \frac{200}{\sqrt{2}} \text{ V}$

(ii) $I_{rms} = \frac{V_{rms}}{R} = \frac{200/\sqrt{2}}{10} = \frac{20}{\sqrt{2}} \text{ A}$

(iii) $P = I_{rms}^2 R = \left(\frac{20}{\sqrt{2}} \right)^2 \times 10 = \frac{400}{2} \times 10 = 2000 \text{ W}$

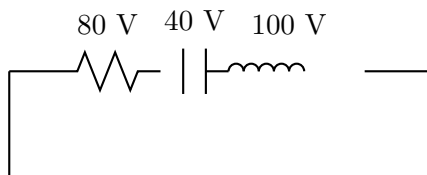
Final Answer

(i) $V_{rms} = \frac{200}{\sqrt{2}} \text{ V} \approx 141.4 \text{ V}$; (ii) $I_{rms} = \frac{20}{\sqrt{2}} \text{ A} \approx 14.1 \text{ A}$; (iii) $P = 2000 \text{ W}$

Question 51

In the given circuit, the potential differences across resistance, capacitance and inductance are given as $V_R = 80 \text{ V}$, $V_C = 40 \text{ V}$, $V_L = 100 \text{ V}$. Work out the emf of the source of alternating current and the power factor of the circuit. (Hint: $\tan \phi = \frac{V_L - V_C}{V_R}$)

★ Likely Exam Question



Series R-C-L circuit with $V_R = 80 \text{ V}$, $V_C = 40 \text{ V}$, $V_L = 100 \text{ V}$

Given: $V_R = 80 \text{ V}$, $V_C = 40 \text{ V}$, $V_L = 100 \text{ V}$

Formula: $V = \sqrt{V_R^2 + (V_L - V_C)^2}$, $\cos \phi = \frac{V_R}{V}$

Solution:

$$V = \sqrt{80^2 + (100 - 40)^2} = \sqrt{6400 + 3600} = \sqrt{10000} = 100 \text{ V}$$

$$\cos \phi = \frac{V_R}{V} = \frac{80}{100} = 0.8$$

Final Answer

$$V = 100 \text{ V}, \quad \cos \phi = 0.8$$

Question 52

An alternating voltage and the corresponding current are given by $V = 110 \sin(\omega t + \pi/6)$ and $I = 5 \sin(\omega t - \pi/6)$ respectively. Find the impedance of the circuit and the average power dissipation in it. (Hint: $\phi = \pi/3$)

Given: $V_0 = 110 \text{ V}$, $I_0 = 5 \text{ A}$, $\phi = \pi/3 = 60^\circ$

Formula: $Z = \frac{V_0}{I_0}$, $P = V_{rms} I_{rms} \cos \phi$

Solution:

$$Z = \frac{110}{5} = 22 \Omega$$

$$P = \frac{V_0}{\sqrt{2}} \times \frac{I_0}{\sqrt{2}} \times \cos \phi = \frac{110}{\sqrt{2}} \times \frac{5}{\sqrt{2}} \times \cos 60^\circ = \frac{110 \times 5}{2} \times 0.5 = 275 \times 0.5 = 137.5 \text{ W}$$

Final Answer

$$Z = 22 \Omega, \quad P = 137.5 \text{ W}$$

Question 53

In an alternating circuit connected to an emf of 100 V and frequency 50 Hz, a resistance of 10Ω and an inductance of $1/(10\pi)$ H are connected in series. Find out the power dissipated in the circuit. (Hint: $P = V_{rms} \times I_{rms} \times \cos \phi$, where $\cos \phi = R/Z$)

Given: $V = 100 \text{ V}$, $f = 50 \text{ Hz}$, $R = 10 \Omega$, $L = \frac{1}{10\pi} \text{ H}$

Formula: $X_L = 2\pi fL$, $Z = \sqrt{R^2 + X_L^2}$, $P = \frac{V^2 R}{Z^2}$

Solution:

$$X_L = 2\pi \times 50 \times \frac{1}{10\pi} = \frac{100\pi}{10\pi} = 10 \Omega$$

$$Z = \sqrt{10^2 + 10^2} = \sqrt{200} = 14.14 \Omega$$

$$I_{rms} = \frac{V}{Z} = \frac{100}{14.14} = 7.07 \text{ A}, \quad \cos \phi = \frac{R}{Z} = \frac{10}{14.14} = 0.707$$

$$P = V_{rms} I_{rms} \cos \phi = 100 \times 7.07 \times 0.707 = 500 \text{ W}$$

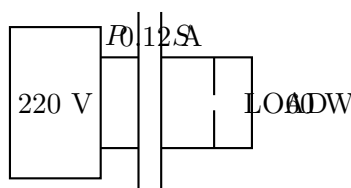
Final Answer

$$P = 500 \text{ W}$$

VIII. Transformer

Question 54

For the given circuit, comment on the type of transformer used. (Primary: 220 V; Secondary current: 0.12 A; Load: 60 W)



Transformer circuit: 220 V primary, secondary feeding a 60 W load at 0.12 A

Given: $V_p = 220 \text{ V}$, $I_s = 0.12 \text{ A}$, $P = 60 \text{ W}$

Formula: $V_s = \frac{P}{I_s}$; if $V_s > V_p$, transformer is step-up; if $V_s < V_p$, step-down

Solution:

$$V_s = \frac{P}{I_s} = \frac{60}{0.12} = 500 \text{ V}$$

Since $V_s = 500 \text{ V} > V_p = 220 \text{ V}$, the secondary voltage is higher than the primary voltage.

Final Answer

Step-up transformer is used.

Question 55

A step-down transformer connected to an AC mains supply of 440 V is made to operate a 22 V, 44 W lamp, ignoring power losses in the transformer. Find the current in the primary circuit and the transformation ratio.

★ Likely Exam Question

Given: $V_p = 440 \text{ V}$, $V_s = 22 \text{ V}$, $P_s = 44 \text{ W}$

Formula: $I_s = \frac{P_s}{V_s}$, $V_p I_p = V_s I_s$ (ideal transformer), $r = \frac{N_s}{N_p} = \frac{V_s}{V_p}$

Solution:

$$I_s = \frac{44}{22} = 2 \text{ A}$$

$$I_p = \frac{V_s I_s}{V_p} = \frac{22 \times 2}{440} = \frac{44}{440} = 0.1 \text{ A}$$

$$r = \frac{V_s}{V_p} = \frac{22}{440} = \frac{1}{20}$$

Final Answer

$$I_p = 0.1 \text{ A}, \quad r = \frac{1}{20}$$

Question 56

A transformer has 1200 turns in the primary and 600 turns in the secondary winding. Primary voltage is 440 V and load in the secondary is 110Ω . Calculate the current in the primary, assuming it to be an ideal transformer.

Given: $N_p = 1200$, $N_s = 600$, $V_p = 440 \text{ V}$, $R_{load} = 110 \Omega$

Formula: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$, $I_s = \frac{V_s}{R_{load}}$, $I_p = I_s \times \frac{N_s}{N_p}$

Solution:

$$V_s = V_p \times \frac{N_s}{N_p} = 440 \times \frac{600}{1200} = 220 \text{ V}$$

$$I_s = \frac{220}{110} = 2 \text{ A}$$

$$I_p = I_s \times \frac{N_s}{N_p} = 2 \times \frac{600}{1200} = 1 \text{ A}$$

Final Answer

$$I_p = 1 \text{ A}$$

Question 57

A power transmission line feeds input power at 2.3 kV to a step-down transformer with its primary winding having 3000 turns. The output power is delivered at 230 V by the transformer. The current in the primary of the transformer is 5 A and its efficiency is 90%. The winding of the transformer is made of copper. Find the output current of the transformer.

★ Likely Exam Question

Given: $V_p = 2.3 \text{ kV} = 2300 \text{ V}$, $I_p = 5 \text{ A}$, $V_s = 230 \text{ V}$, $\eta = 90\%$

Formula: $P_{in} = V_p I_p$, $P_{out} = \eta \times P_{in}$, $I_s = \frac{P_{out}}{V_s}$

Solution:

$$P_{in} = V_p I_p = 2300 \times 5 = 11500 \text{ W}$$

$$P_{out} = 0.90 \times 11500 = 10350 \text{ W}$$

$$I_s = \frac{P_{out}}{V_s} = \frac{10350}{230} = 45 \text{ A}$$

Final Answer

$$I_s = 45 \text{ A}$$

Common Mistake: Since efficiency is less than 100%, output power must be found using $P_{out} = \eta \times P_{in}$, not simply $P_{in} = P_{out}$ as for an ideal transformer.